

Source-aware machine learning for heavy-metal adsorption: high-accuracy interpolation versus cross-source generalization

Yash Bisht^{a,*}, Susmit Chitransh^b

^a*Department of Mechanical and Industrial Engineering, Indian Institute of Technology Roorkee, Roorkee 247667, Uttarakhand, India*

^b*Department of Chemical Engineering, University of Cape Town, Rondebosch 7700, Cape Town, South Africa*

Abstract

Machine learning is increasingly used to predict adsorption removal efficiency for water and wastewater treatment, but reported performance can depend strongly on feature definition and validation design. This study develops a source-aware and leakage-aware framework for heavy-metal adsorption removal-efficiency prediction using a literature-compiled experimental dataset. The raw dataset contained 566 adsorption records; after deterministic numeric parsing and quality filtering, 563 valid records remained, covering 189 normalized source references, 219 adsorbents, and 16 normalized adsorbate classes. Four feature sets were evaluated to separate capacity-inclusive process-monitoring models from capacity-free pre-experiment screening models. Two validation tracks were used. Track A quantified source-specific interpolation through within-source cross-validation, where XGBoost models reached R^2 values of 0.992, 0.974, and 0.960 for three major source groups. Track B evaluated broader generalization through repeated random row-wise, adsorbent-held-out, and reference-held-out splits. The full capacity-inclusive XGBoost model achieved $R^2 = 0.798$ under random row-wise validation and $R^2 = 0.577$ under adsorbent-held-out validation. In contrast, reference-held-out validation reduced the best observed screening-model performance to $R^2 = 0.112$. These results show that high adsorption-prediction accuracy is scientifically meaningful for in-source interpolation, but not sufficient

*Corresponding author.

evidence of universal deployment. The proposed dual-track protocol provides a practical reporting framework for adsorption machine-learning studies by distinguishing interpolation accuracy, process-monitoring prediction, pre-experiment screening, and cross-source generalization risk.

Keywords: adsorption, heavy metals, removal efficiency, machine learning, XGBoost, data leakage, source-aware validation, wastewater treatment

Highlights

- Source-specific adsorption models reached removal-efficiency prediction R^2 up to 0.992.
- Random row-wise validation overstated performance relative to source-held-out testing.
- Reference-held-out validation reduced the best observed R^2 to approximately 0.11.
- Capacity-inclusive and capacity-free models answer different adsorption-design questions.
- A dual-track validation protocol is proposed for adsorption machine-learning studies.

1. Introduction

Heavy-metal contamination remains a persistent challenge for water and wastewater treatment because metals such as Pb, Cd, Cr, Ni, Cu, As, Hg, and Zn are toxic, non-biodegradable, and prone to accumulation in environmental and biological systems. Adsorption is widely investigated for heavy-metal removal because it can operate under mild conditions, can be adapted to low-cost or waste-derived adsorbents, and can be optimized through pH, adsorbent dose, contact time, temperature, surface chemistry, and initial contaminant concentration. In parallel, the growth of experimental adsorption datasets has encouraged the use of machine learning (ML) to predict removal efficiency and guide adsorbent selection.

Recent adsorption and environmental-remediation ML studies demonstrate that high predictive accuracy is possible. Hafsa et al. modeled heavy-metal adsorption efficiencies using multiple ML algorithms and reported R^2

values in the 0.96–0.99 range under ten-fold cross-validation, with experimental data supplemented by synthetic interpolation [1]. Lu et al. predicted heavy-metal removal by chitosan-based flocculants using random forests and reported $R^2 = 0.9354$, identifying solution pH and flocculant molecular weight as important drivers [2]. Wang et al. used interpretable ML for heavy-metal removal by biochar and reported XGBoost test performance near $R^2 = 0.99$, together with SHAP and partial-dependence analyses to relate removal efficiency to biochar and operating variables [3, 6]. Recent ensemble-learning work on biochar adsorption similarly reported strong XGBoost performance for heavy-metal adsorption efficiency prediction [4]. Within Journal of Environmental Chemical Engineering, interpretable XGBoost modeling has also been used for heavy-metal removal efficiency in electrokinetic remediation, with SHAP-based interpretation supporting process understanding [5]. These studies collectively show that nonlinear tree-based models can learn useful structure-property-process relationships in environmental treatment datasets.

However, high row-wise validation performance does not necessarily imply generalization to unseen laboratories, adsorbents, pollutant systems, or source studies. In ML-based science, data leakage and validation mismatch can produce overoptimistic estimates of deployable performance [7]. Environmental ML best-practice guidance similarly emphasizes careful data splitting, leakage management, randomness assessment, and model evaluation design [8]. These concerns are especially relevant for heterogeneous adsorption datasets because different sources may use different adsorbent synthesis routes, pollutant species, concentration ranges, analytical protocols, and reporting conventions.

Feature definition is another important source of ambiguity. Adsorption capacity and removal efficiency are both derived from concentration changes in many batch adsorption experiments:

$$\text{RE}(\%) = 100 \frac{C_0 - C_e}{C_0}, \quad (1)$$

$$q_e = \frac{(C_0 - C_e)V}{m}, \quad (2)$$

where C_0 and C_e are the initial and equilibrium concentrations, V is solution volume, and m is adsorbent mass. Capacity can therefore be useful for process monitoring after adsorption has been measured, but it should be

treated carefully in pre-experiment screening models where capacity is not yet known. Recent PFAS adsorption modeling has also shown that random-split performance can be much stronger than leave-one-compound-out performance, illustrating that interpolation and transfer to unseen chemical domains are distinct tasks [9].

This study addresses these issues by developing a dual-track validation framework for heavy-metal adsorption removal-efficiency prediction. Track A evaluates source-specific interpolation and asks whether high R^2 values comparable to the adsorption-ML literature can be reproduced within individual source groups. Track B evaluates global generalization under random row-wise, adsorbent-held-out, and reference-held-out validation. The study further separates capacity-inclusive process-monitoring models from capacity-free screening models. The objective is not only to maximize R^2 , but to identify when high R^2 is scientifically valid, when it is validation-dependent, and how adsorption ML studies should report performance for practical use.

2. Materials and methods

2.1. Dataset

The dataset was compiled from reported heavy-metal adsorption experiments. Each record corresponds to one adsorption condition with a reported removal efficiency. The raw dataset contained 566 rows and 13 columns. After numeric parsing, missing-value filtering, and constraining removal efficiency to the physically meaningful range 0–100%, 563 records remained. The cleaned analysis dataset included 189 normalized source references, 219 adsorbents, and 16 normalized adsorbate classes.

The target variable was removal efficiency, RE (%). Input variables included adsorbent name, adsorbate identity, surface area, adsorption capacity, initial concentration, contact time, adsorbent dose, agitation speed, initial pH, temperature, and source reference. The source identifier, denoted **Ref_group**, was used as a proxy for source origin because the present dataset does not yet contain explicit laboratory-origin labels for all records.

2.2. Numeric cleaning and source normalization

Literature-derived adsorption tables often contain nonuniform numeric formatting. Numeric entries were cleaned using deterministic rules: decimal commas were converted to decimal points, uncertainty and approximate

symbols were removed, multiple values in one cell were reduced to the first numeric value, and nonnumeric text was parsed using regular expressions. Rows with missing values in required numeric variables were removed. Adsorbate names were normalized to reduce duplicates such as elemental and valence-state aliases. Adsorbent names were grouped into broad families, including carbon-based, mineral/oxide, polymer, biomass, and composite/other categories.

The cleaned data were assigned to four pH-concentration regimes using $\text{pH} = 7$ and the median initial concentration, $C_0 = 50$ ppm (Table 1). The relatively small size of the neutral/low-concentration regime was considered when interpreting regime-wise results.

Table 1: pH-concentration regime counts in the cleaned dataset.

Regime	Count
Acidic_HighC0	246
Acidic_LowC0	231
Neutral_HighC0	60
Neutral_LowC0	26

2.3. Feature engineering

Four feature sets were constructed to distinguish scientific use cases.

Full capacity process model. This feature set included operating variables, surface area, adsorption capacity, capacity-derived ratios, ion descriptors, and adsorbent/adsorbate categories. It is suitable for descriptive modeling or process monitoring when capacity is known.

Screening-safe plus ion model. This feature set excluded adsorption capacity and capacity-derived terms. It retained operating variables, surface area, physics-inspired transformations, ion descriptors, and adsorbent family.

Screening-safe plus adsorbate one-hot model. This feature set excluded adsorption capacity and used adsorbate identity as a categorical variable.

Screening-safe numeric-only model. This was the strictest feature set, excluding adsorption capacity, capacity-derived variables, and adsorbate identity.

Physics-inspired variables included site density, logarithmic contact time, mixing index, driving-force proxy, and acidity strength:

$$\text{Site Density} = \frac{\text{dose} \times \text{surface area}}{C_0}, \quad (3)$$

$$\text{LogTime} = \log(1 + \text{contact time}), \quad (4)$$

$$\text{Mixing Index} = \text{RPM} \times \text{LogTime}, \quad (5)$$

$$\text{Driving Force} = \frac{C_0}{\text{dose}}, \quad (6)$$

$$\text{Acidity Strength} = |\text{pH} - 7|. \quad (7)$$

Ion descriptors included charge, ionic radius, hydrated radius, ionic potential, and hydration ratio. These descriptors were used to reduce reliance on one-hot pollutant labels.

2.4. Models

125 XGBoost was selected as the primary model because gradient-boosted trees are effective for nonlinear tabular datasets and are widely used in adsorption ML studies [10, 11]. Extremely randomized trees (ExtraTrees) were included as a comparator ensemble model because random forest-style tree ensembles provide robust nonlinear baselines for tabular prediction [12, 13].
 130 A mean dummy regressor was included to quantify improvement beyond a no-skill baseline. This study reports XGBoost and ExtraTrees because they capture the main performance trends while keeping the analysis reproducible and computationally light.

2.5. Validation protocols

135 Two validation tracks were used.

Track A: source-specific interpolation. For each source/reference with at least 20 records, source-specific 5-fold cross-validation was performed. This tests whether high R^2 values are achievable within a known experimental source, which is similar to the validation setting of many high-performance
 140 adsorption ML studies.

Track B: global generalization. Three global validation protocols were applied:

1. Random row-wise split stratified by pH-concentration regime.
2. Group split holding out adsorbent names.
- 145 3. Group split holding out entire references/sources.

Each Track B split was repeated over five random seeds. Metrics included R^2 , root mean squared error (RMSE), mean absolute error (MAE), and bias. Reference-held-out validation was treated as the most conservative source-transfer test available without explicit laboratory-origin labels.

150 3. Results

3.1. Source-specific models reproduce high- R^2 adsorption prediction

Track A showed that high removal-efficiency prediction accuracy is achievable when models are trained and evaluated within individual source groups. The full capacity process model achieved $R^2 = 0.9916$ for source group Shen et al. (2017b), $R^2 = 0.9735$ for Gao et al. (2019), and $R^2 = 0.9601$ for Zama et al. (2017). These results are in the range commonly reported in adsorption ML studies under row-wise or within-domain validation [1, 3, 2].

Table 2: Track A source-specific 5-fold cross-validation results for the full capacity process model.

Source group	Rows	R^2	RMSE	MAE
Shen et al. (2017b)	65	0.9916	3.20	2.33
Gao et al. (2019)	48	0.9735	2.97	2.14
Zama et al. (2017)	91	0.9601	7.84	3.56
Cui et al. (2016a)	24	0.8936	6.65	4.56

Capacity-free source-specific models also performed well in some cases. For example, the source-specific model for Zama et al. (2017) achieved $R^2 = 0.9250$ using the screening-safe ion feature set, while Shen et al. (2017b) achieved R^2 values above 0.91 using capacity-free screening feature sets. This indicates that high source-specific performance is not solely a capacity-leakage artifact, although capacity-inclusive models generally gave the strongest interpolation performance.

165 3.2. Global row-wise validation gives moderate-to-good performance

Under repeated random row-wise splits, the full capacity process model achieved $R^2 = 0.7975$ and RMSE = 13.16. The screening-safe plus ion model achieved $R^2 = 0.7278$ and RMSE = 15.26, while the strict numeric-only screening model achieved $R^2 = 0.6863$ and RMSE = 16.43. Thus, even without adsorption capacity, the model retained useful predictive ability under in-dataset interpolation (Table 3).

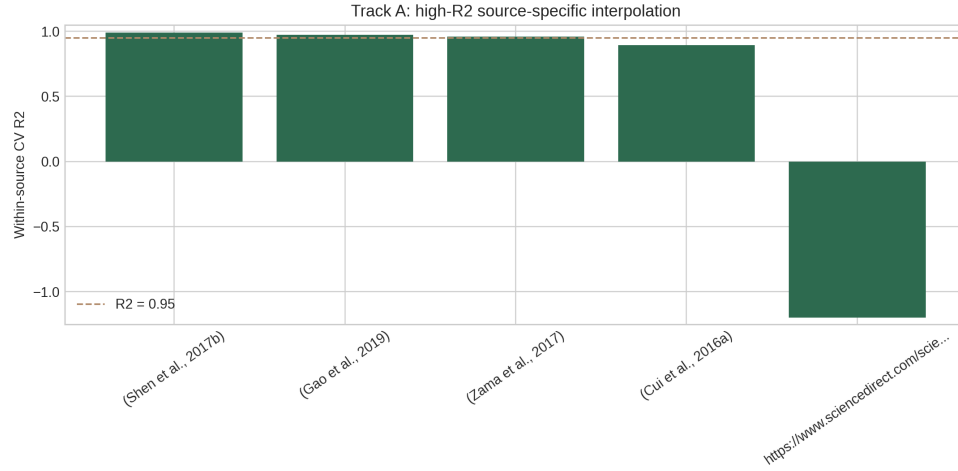


Figure 1: Track A source-specific interpolation performance. Several individual source-group models achieved R^2 values near or above 0.95, while other source groups remained more difficult.

Table 3: Random row/regime split performance. Values are means over five repeated splits.

Feature/model	R^2	RMSE	MAE
Full capacity XGBoost	0.7975	13.16	8.32
Screening-safe plus ion XGBoost	0.7278	15.26	10.80
Screening-safe numeric-only XGBoost	0.6863	16.43	12.03

These results support the use of ML for in-domain adsorption prediction. However, random row-wise validation alone is not sufficient evidence for deployment on unseen sources.

175 3.3. Generalization weakens under adsorbent- and reference-held-out validation

The validation protocol strongly affected performance. When entire adsorbents were held out, the full capacity process model achieved $R^2 = 0.5768$ and RMSE = 16.38. The safer numeric-only screening model achieved $R^2 =$
180 0.4886 and RMSE = 18.49. This indicates moderate but imperfect transfer to unseen adsorbent names.

Reference-held-out validation was substantially more difficult. The best reference-held-out result was obtained by the screening-safe numeric-only ExtraTrees model, with $R^2 = 0.1117$ and RMSE = 26.68. XGBoost achieved
185 R^2 values near or below zero under reference-held-out validation, including $R^2 = -0.0553$ for the full capacity process model and $R^2 = -0.0223$ for the screening-safe numeric-only model.

Table 4: Track B global generalization performance under stricter group-held-out validation. Values are means over five repeated splits.

Validation		Feature/model	R^2	RMSE	MAE
Holdout adsorbent		Full capacity XGBoost	0.5768	16.38	11.18
Holdout adsorbent		Screening-safe numeric-only XGBoost	0.4886	18.49	13.71
Holdout	refer-	Screening-safe numeric-only	0.1117	26.68	20.43
ence/source		ExtraTrees			
Holdout	refer-	Screening-safe numeric-only	-0.0223	28.42	21.86
ence/source		XGBoost			
Holdout	refer-	Full capacity XGBoost	-0.0553	28.34	21.84
ence/source					

These results show that removal-efficiency prediction is strongly source-dependent. The same data and model family can produce high R^2 under
190 source-specific interpolation, moderate R^2 under random global validation, and weak R^2 under source-held-out validation.

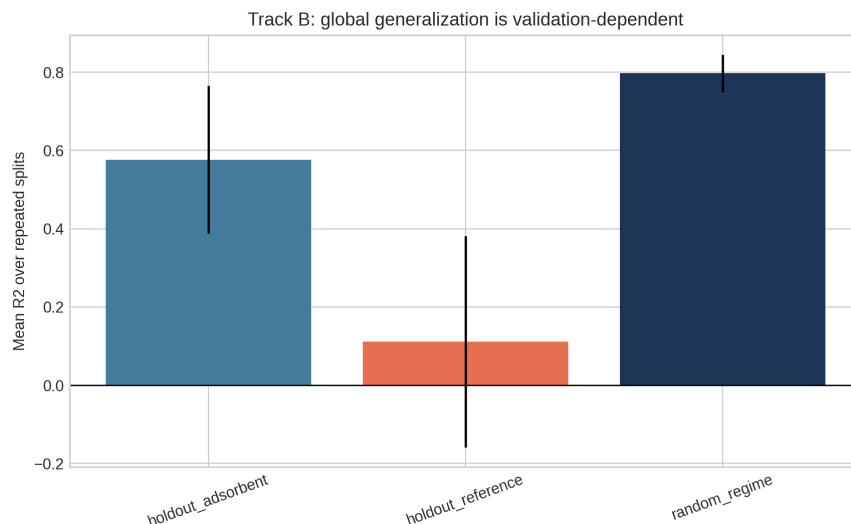


Figure 2: Track B validation comparison. Row-wise validation, adsorbent-held-out validation, and reference-held-out validation represent increasingly strict generalization tasks.

3.4. Capacity-inclusive and capacity-free models answer different questions

The full capacity process model consistently performed best for random row-wise validation and several source-specific models. This is expected because adsorption capacity carries strong information about adsorption performance. However, capacity can be outcome-adjacent to RE and may not be available before an experiment. Therefore, capacity-inclusive models are best framed as process-monitoring or descriptive models. Capacity-free models are more appropriate for pre-experiment screening and adsorbent-design guidance, even though their performance is lower.

4. Discussion

4.1. Why near-0.95 R^2 is both achievable and limited

The Track A results explain why many adsorption ML studies report R^2 values near or above 0.95. Within a single source, adsorbent family, pollutant system, or experimental protocol, the response surface can be learned accurately by tree-based models. In this setting, high R^2 is not surprising and is not necessarily invalid. The present results reproduce that behavior, with source-specific R^2 values of 0.960–0.992 for three major source groups.

However, Track B shows that this performance should not be interpreted
 210 as universal adsorption prediction. When sources are held out, the model
 must extrapolate across experimental protocols, adsorbent synthesis meth-
 ods, pollutant distributions, concentration ranges, and reporting conventions.
 Under that validation setting, performance fell sharply. This finding is con-
 sistent with recent adsorption-related work showing that random-split per-
 215 formance can be substantially higher than leave-compound-out performance
 [9].

4.2. Relation to existing adsorption ML literature

The proposed framework complements rather than contradicts previous
 high-accuracy adsorption ML studies. Hafsa et al. demonstrated that tree-
 220 based algorithms can achieve high removal-efficiency prediction accuracy for
 heavy-metal adsorption datasets under repeated cross-validation [1]. Wang
 et al. showed that XGBoost, SHAP, and partial-dependence analyses can
 identify interpretable drivers of heavy-metal removal by biochar [3]. Lu et
 al. showed that random forests can predict heavy-metal removal by chitosan-
 225 based flocculants with R^2 above 0.93 [2]. Barkhordari et al. further demon-
 strated that interpretable XGBoost models are relevant to JECE’s environ-
 mental remediation scope by predicting heavy-metal removal efficiency in
 electrokinetic remediation [5]. These studies establish that ML is useful for
 environmental treatment modeling.

230 The present study adds a validation-layer contribution: adsorption ML
 workflows should explicitly state whether they are predicting within a known
 source or generalizing to unseen sources. This distinction is critical because
 high row-wise R^2 can be overoptimistic when the deployment target is a new
 laboratory, adsorbent synthesis route, pollutant matrix, or literature source.
 235 The broader ML literature has identified leakage and validation mismatch
 as major reproducibility risks [7, 8], and adsorption datasets have several
 mechanisms by which such mismatch can occur.

4.3. Practical implications

For practical adsorption screening, a single universal RE model should
 240 be used cautiously. A more reliable workflow is:

1. Use source-specific or laboratory-calibrated models for high-accuracy
 optimization within a known experimental system.

2. Use capacity-free screening models for preliminary adsorbent and operating-condition selection.
- 245 3. Use source-held-out validation to estimate external deployment risk.
4. Report applicability-domain or uncertainty information when making predictions for new adsorbents, pollutants, or laboratories.

This framing makes high- R^2 models useful without overstating their generality.

250 4.4. *Limitations and future work*

Several limitations should be addressed before final journal submission. First, the current dataset does not contain explicit laboratory-origin labels for all records; bibliographic reference was therefore used as a source proxy. Future versions should annotate records as in-house, external-laboratory, or literature-derived and repeat validation with leave-laboratory-out splits. Second, the ion descriptor table should be expanded with fully cited descriptors such as electronegativity, hydration energy, softness/hardness, and likely aqueous speciation. Third, key adsorbent descriptors such as pH_{pzc} , pore volume, pore size, zeta potential, FTIR-derived functional groups, and cation-exchange capacity are incomplete. Fourth, the neutral/low-concentration regime is underrepresented. Finally, reference-held-out performance is currently weak, so the model should not be presented as a universal RE predictor.

5. Conclusions

265 This study developed a source-aware and leakage-aware framework for adsorption removal-efficiency prediction. The results show that high R^2 values are achievable within individual adsorption sources, with source-specific XGBoost models reaching R^2 up to 0.992. However, global performance depends strongly on the validation protocol: random row-wise validation reached $R^2 = 0.798$ for the full capacity process model, while reference-held-out validation reduced the best observed R^2 to approximately 0.112. These findings show that high-accuracy interpolation and cross-source generalization are different scientific tasks. Future adsorption ML studies should report both, separate capacity-inclusive process-monitoring models from capacity-free screening models, and include source- or laboratory-held-out validation when deployment beyond the training source is claimed.

CRedit authorship contribution statement

Yash Bisht: Conceptualization, Methodology, Software, Formal analysis, Visualization, Writing – original draft. Susmit Chitransh: Data curation, Investigation, Validation, Writing – review and editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Data availability

The curated dataset, analysis scripts, and generated benchmark outputs are available from the corresponding author upon reasonable request and will be deposited in a public repository before final publication.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used OpenAI’s Codex to assist with code organization, LaTeX formatting, and language polishing. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the manuscript.

Acknowledgements

The authors acknowledge the institutional support of the Indian Institute of Technology Roorkee and the University of Cape Town.

References

- [1] N. Hafsa, S. Rushd, M. Al-Yaari, M. Rahman, A generalized method for modeling the adsorption of heavy metals with machine learning algorithms, *Water* 12 (2020) 3490. <https://doi.org/10.3390/w12123490>.
- 305 [2] C. Lu, Z. Xu, B. Dong, Y. Zhang, M. Wang, Y. Zeng, C. Zhang, Machine learning for the prediction of heavy metal removal by chitosan-based flocculants, *Carbohydrate Polymers* 285 (2022) 119240. <https://doi.org/10.1016/j.carbpol.2022.119240>.
- 310 [3] C. Wang, Y. Zhao, Y. Gao, H. Chen, X. Li, B. Zhou, D. Fan, Z. Fang, J. Liu, Interpretable machine learning for predicting heavy metal removal and optimizing biochar characteristics, *Journal of Water Process Engineering* 68 (2024) 106484. <https://doi.org/10.1016/j.jwpe.2024.106484>.
- 315 [4] Z.M. Yaseen, F.L. Alhalimi, Heavy metal adsorption efficiency prediction using biochar properties: a comparative analysis for ensemble machine learning models, *Scientific Reports* 15 (2025). <https://doi.org/10.1038/s41598-025-96271-5>.
- 320 [5] M.S. Barkhordari, N. Zhou, K. Li, C. Qi, Interpretable machine learning for predicting heavy metal removal efficiency in electrokinetic soil remediation, *Journal of Environmental Chemical Engineering* 12 (2024) 114330. <https://doi.org/10.1016/j.jece.2024.114330>.
- [6] S.M. Lundberg, S.-I. Lee, A unified approach to interpreting model predictions, in: *Advances in Neural Information Processing Systems* 30, 2017, pp. 4765–4774.
- 325 [7] S. Kapoor, A. Narayanan, Leakage and the reproducibility crisis in machine-learning-based science, *Patterns* 4 (2023) 100804. <https://doi.org/10.1016/j.patter.2023.100804>.
- 330 [8] J.-J. Zhu, M. Yang, Z.J. Ren, Machine learning in environmental research: Common pitfalls and best practices, *Environmental Science & Technology* 57 (2023) 17671–17689. <https://doi.org/10.1021/acs.est.3c00026>.

- 335 [9] H.V. Patel, J. Green, H. Park, S. Luster-Teasley Pass, R. Zhao, A hybrid response surface methodology and machine learning framework for quantifying effects of physicochemical parameters on PFAS distribution, ACS ES&T Water (2026) Advance online publication. <https://doi.org/10.1021/acsestwater.5c01162>.
- 340 [10] T. Chen, C. Guestrin, XGBoost: A scalable tree boosting system, in: Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, ACM, San Francisco, CA, USA, 2016, pp. 785–794. <https://doi.org/10.1145/2939672.2939785>.
- [11] J.H. Friedman, Greedy function approximation: A gradient boosting machine, *Annals of Statistics* 29 (2001) 1189–1232. <https://doi.org/10.1214/aos/1013203451>.
- 345 [12] L. Breiman, Random forests, *Machine Learning* 45 (2001) 5–32. <https://doi.org/10.1023/A:1010933404324>.
- [13] P. Geurts, D. Ernst, L. Wehenkel, Extremely randomized trees, *Machine Learning* 63 (2006) 3–42. <https://doi.org/10.1007/s10994-006-6226-1>.